Non-Functional Requirements Specification (NFRS)

Phase 2: Performance, Quality, and Governance Standards

Project: Global Olympic Sports Performance Analysis • **Author:** Md. Firdous Alam • **Era:** 1896–2016 Centennial Dataset

1. Non-Functional Criteria & Verifications

Non-functional constraints enforce enterprise reliability and GitHub deployment compatibility.

- Storage Compliance: The primary dataset athletes.csv must remain strictly below GitHub's 25 MB web upload limit. Actual size: 22,788,167 bytes (21.73 MB).
- Processing Speed: The complete Pandas data processing pipeline must execute in under 5.0 seconds on standard dual-core hardware. Actual benchmark: 1.84 seconds.
- Integrity: Zero duplicate composite keys [games_id, noc] in medal_results.csv. Actual duplicates: 0 across 3,837 rows.
- Usability: Tableau dashboards must render cleanly on both desktop (1920x1080) and laptop displays.